

Foundations of Twin Quantum Geometry: Emergent Quantization from Discrete Orbital Structures

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Abstract

We develop a geometric foundation for quantum mechanics based on discrete group actions on phase space, formalized as twin spaces. Starting from a classical symplectic phase space (M, ω) and a discrete twin group G_{disc} acting on M , we show that the Planck constant $\hbar$, canonical commutation relations, the Hilbert space structure and Schrödinger dynamics can be derived from orbital geometry rather than postulated. The Planck constant appears as the minimal non-trivial symplectic area of a fundamental twin cell, or equivalently as a cohomological invariant of the group action. Canonical commutation relations emerge from a non-trivial braiding of position and momentum translations on the twin lattice. We construct a natural twin Hilbert space $\mathcal{H}_{\text{twin}} = L^2(M/G_{\text{disc}}, \mu)$ and prove that Schrödinger dynamics arises as covariant transport of Hamiltonian flows on the orbit space, under a quantization condition for the symplectic area. Several model systems (harmonic oscillator, free particle, particle on a circle) illustrate how standard quantum spectra are recovered within this purely geometric framework.

Keywords. Twin spaces, discrete group actions, symplectic geometry, emergent quantization, canonical commutation relations, Hilbert space, Schrödinger equation.

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1 Introduction

The standard axioms of quantum mechanics postulate the existence of

- a complex Hilbert space $\mathcal{H}$ of states;
- a distinguished constant $\hbar > 0$;
- self-adjoint operators $Q, P, \dots$ satisfying canonical commutation relations, such as $[Q, P] = i\hbar \text{Id}$;
- the Born rule, prescribing probabilities as squared norms of orthogonal projections.

In this picture, the Hilbert space, the value of $\hbar$, and the canonical structure are given *a priori*. Even in geometric quantization and deformation quantization, quantization is encoded in additional structures rather than emerging from a more primitive geometry.

In this article we propose a different viewpoint. We start from a classical symplectic phase space (M, ω) and a discrete group G_{disc} acting on M by symplectomorphisms, hence generating a lattice-like structure of orbits. We show that, under suitable assumptions, the following objects emerge from this orbital geometry:

- a distinguished area scale which we identify with $\hbar$;
- canonical commutation relations derived from the braiding of discrete position and momentum translations;
- a natural Hilbert space $\mathcal{H}_{\text{twin}} = L^2(M/G_{\text{disc}}, \mu)$ associated with the orbit space and an invariant measure μ ;
- a unitary time evolution on $\mathcal{H}_{\text{twin}}$ satisfying a Schrödinger-type equation arising from covariant transport of classical Hamiltonian flows.

This leads to the following main results.

Theorem 1.1 (Emergent Planck constant). *Let (M, ω) be a $2n$ -dimensional symplectic manifold and let G_{disc} be a discrete group of symplectomorphisms acting properly and cocompactly on M . Assume that the symplectic form has integral periods on fundamental homology cycles associated with the action. Then there exists a constant $\hbar > 0$ such that the symplectic area of any fundamental cell of the action is an integer multiple of $2\pi\hbar$. The minimal non-zero area*

$$\Delta\mathcal{A}_{\min} = 2\pi\hbar$$

is uniquely determined by the cohomology class $[\omega]$ and the action of G_{disc} .

Theorem 1.2 (Geometric canonical commutation). *Let $T_q(a)$ and $T_p(b)$ denote discrete translations in position and momentum associated with generators of G_{disc} along canonical directions in $M = \mathbb{R}^2$ or $M = \mathbb{R}^{2n}$. Under the quantization condition of Theorem 1.1 one has*

$$T_q(a)T_p(b) = e^{iab/\hbar}T_p(b)T_q(a).$$

In the infinitesimal limit $a, b \rightarrow 0$ this yields the canonical commutation relation

$$[Q, P] = i\hbar\text{Id}$$

for densely defined self-adjoint generators Q, P of these translations on a suitable Hilbert space.

Theorem 1.3 (Twin Hilbert construction). *Assume that G_{disc} acts properly discontinuously on M and that the quotient $X := M/G_{\text{disc}}$ carries a σ -finite measure μ invariant under the induced dynamics. Then the space*

$$\mathcal{H}_{\text{twin}} := L^2(X, \mu)$$

is a separable Hilbert space, and the representations of translations, observables and Hamiltonian flows define unitary (or self-adjoint) operators on $\mathcal{H}_{\text{twin}}$ compatible with the twin geometry.

Theorem 1.4 (Emergent Schrödinger dynamics). *Let $H_{\text{cl}} : M \rightarrow \mathbb{R}$ be a classical Hamiltonian whose flow preserves the quantized symplectic structure and commutes with the action of G_{disc} up to gauge-equivalent phases. Then the induced evolution on $\mathcal{H}_{\text{twin}}$ is unitary and satisfies a Schrödinger equation*

$$i\hbar \frac{\partial}{\partial t} \psi_t = \hat{H} \psi_t,$$

where $\hat{H}$ is the twin-quantized Hamiltonian acting on $\mathcal{H}_{\text{twin}}$.

The present article provides a detailed construction and proof of these results, together with illustrative examples. It serves as the geometric and analytic foundation for subsequent works:

- in *Twin Quantum Mechanics: Operators, Measurement and Entanglement*, we develop the full operator, measurement and entanglement theory on $\mathcal{H}_{\text{twin}}$;
- in *Twin Quantum Fields, Kernel Dynamics and Gravity-like Effects*, we extend the construction to quantum fields, kernel dynamics and emergent gravity-like corrections.

Structure of the paper

In Section 2 we introduce twin phase spaces and discrete group actions. Section 3 derives the Planck constant from orbital geometry. In Section 4 we construct geometric canonical commutation relations from the braiding of position and momentum translations. Section 5 defines the twin Hilbert space as L^2 on the orbit space. Section 6 derives Schrödinger dynamics from covariant transport of Hamiltonian flows. Finally, Section 7 presents simple models.

2 Twin phase spaces and discrete group actions

We start from a classical phase space and introduce its twin structure via a discrete group of symplectomorphisms.

2.1 Classical phase space

Let (M, ω) be a $2n$ -dimensional symplectic manifold, where ω is a closed non-degenerate 2-form. Typical examples include:

- $M = \mathbb{R}^{2n}$ with canonical coordinates (q, p) and $\omega = \sum_{j=1}^n dp_j \wedge dq_j$;
- the cotangent bundle $M = T^*Q$ of a configuration manifold Q , with its canonical symplectic form;
- compact phase spaces such as $T^*\mathbb{T}^n$ or coadjoint orbits.

2.2 Discrete twin groups and orbits

Definition 2.1 (Discrete twin group). A *discrete twin group* on (M, ω) is a countable group G_{disc} together with an injective morphism

$$\rho : G_{\text{disc}} \rightarrow \text{Symp}(M, \omega),$$

where $\text{Symp}(M, \omega)$ denotes the group of symplectomorphisms of (M, ω) , such that $\rho(G_{\text{disc}})$ acts properly discontinuously on M .

We write $g \cdot x$ for the action of $g \in G_{\text{disc}}$ on $x \in M$. For each $x \in M$, its orbit is

$$\mathcal{O}_x := \{g \cdot x : g \in G_{\text{disc}}\}.$$

Definition 2.2 (Twin phase space). The pair (M, G_{disc}) , together with the symplectic form ω , is called a *twin phase space*. The orbit space

$$X := M/G_{\text{disc}}$$

is the *twin orbit space*. Points in X correspond to orbits, or equivalently to equivalence classes $[x] = \mathcal{O}_x$ under the action of G_{disc} .

We typically assume that the action is free or has mild stabilizers so that X carries a reasonable differential or measurable structure.

2.3 Fundamental domains and cells

Under suitable assumptions, one can choose a fundamental domain for the action.

Definition 2.3 (Fundamental domain). A measurable subset $F \subset M$ is a *fundamental domain* for the action of G_{disc} if

- $M = \bigsqcup_{g \in G_{\text{disc}}} g \cdot F$ up to measure zero;
- $g \cdot F$ and $h \cdot F$ are disjoint up to measure zero if $g \neq h$.

In the examples of interest, F can be chosen to have finite symplectic volume

$$\text{Vol}_\omega(F) := \int_F \omega^{\wedge n} < \infty.$$

Remark 2.4 (Twin cells). We interpret F (or its translates) as fundamental *twin cells* in phase space. In dimension 2, these are area cells in the (q, p) plane; in higher dimension they are $2n$ -dimensional volume cells. The minimal non-zero symplectic volume of such cells will be related to the Planck constant.

3 Emergence of the Planck constant from orbital geometry

We now show how a distinguished area scale arises from the symplectic structure and the discrete action.

3.1 Integral symplectic periods

Consider the de Rham cohomology class $[\omega] \in H_{\text{dR}}^2(M)$ and the image of integral homology cycles of the quotient. Let

$$\pi : M \rightarrow X = M/G_{\text{disc}}$$

denote the quotient map. Closed 2-cycles Σ in M induced by fundamental loops of the action may carry non-zero symplectic area

$$\int_\Sigma \omega.$$

Definition 3.1 (Quantized symplectic action). We say that the symplectic form is *quantized with respect to G_{disc}* if there exists a constant $\hbar > 0$ such that

$$\int_\Sigma \omega \in 2\pi\hbar\mathbb{Z}$$

for all 2-cycles Σ generated by the action of G_{disc} .

Remark 3.2. This condition is reminiscent of the prequantization condition in geometric quantization, but here it is imposed relative to the discrete group action rather than to all cycles in $H_2(M, \mathbb{Z})$.

3.2 Emergent minimal area

Under this assumption, we can define the minimal non-zero area.

Proposition 3.3 (Minimal symplectic area). *Assume that the set of non-zero symplectic areas*

$$\mathcal{S} := \left\{ \int_{\Sigma} \omega : \Sigma \text{ is a 2-cycle generated by } G_{\text{disc}} \right\}$$

is a discrete subset of $\mathbb{R}$. Then there exists a minimal non-zero value $\Delta\mathcal{A}_{\min} > 0$ in $\mathcal{S}$.

Proof. If $\mathcal{S}$ is discrete and contains non-zero elements, there is a smallest positive element by the well-ordering of discrete subsets of $\mathbb{R}_{\geq 0}$. $\square$

Theorem 3.4 (Emergent Planck constant). *Under the quantization and discreteness assumptions above, there exists $\hbar > 0$ such that every fundamental 2-cycle Σ generated by G_{disc} satisfies*

$$\int_{\Sigma} \omega \in 2\pi\hbar\mathbb{Z},$$

and the minimal non-zero area is

$$\Delta\mathcal{A}_{\min} = 2\pi\hbar.$$

The constant $\hbar$ is uniquely determined by the cohomology class $[\omega]$ restricted to the subgroup generated by the action of G_{disc} .

Idea of the proof. The action of G_{disc} generates a subgroup Γ of $H_2(M, \mathbb{Z})$ spanned by fundamental 2-cycles associated with the group action (for instance, rectangles in (q, p) -coordinates for translations). The quantization condition implies that $\int_{\Sigma} \omega$ lies in a discrete subgroup of $\mathbb{R}$, isomorphic to $\mathbb{Z}$ times a fixed generator. Define $\hbar$ by requiring that the generator equals $2\pi\hbar$. Uniqueness follows from the fact that rescaling $\hbar$ would change the quantized values. $\square$

Remark 3.5 (Cohomological interpretation). On the quantized cycles themselves $\int_{\Sigma} \omega \in 2\pi\hbar\mathbb{Z}$, so $\exp(i \int_{\Sigma} \omega / \hbar) = 1$: the cocycle is *trivial* on the lattice, which is exactly why the lattice translations commute (Proposition 4.1). The non-trivial structure lives one level up—the continuous Heisenberg group of sub-cell displacements carries a non-trivial central extension, a class in H^2 normalized by $\hbar$ —and the integrality condition is precisely the statement that this class restricts trivially to the discrete subgroup G_{disc} .

4 Geometric canonical commutation relations

We now derive canonical commutation relations from the braiding of discrete translations in position and momentum.

4.1 Discrete translations in phase space

For simplicity, consider $M = \mathbb{R}^2$ with coordinates (q, p) and $\omega = dp \wedge dq$. Let $a, b > 0$ and define translations

$$\tau_q(a) : (q, p) \mapsto (q + a, p), \quad \tau_p(b) : (q, p) \mapsto (q, p + b).$$

Assume that these generate a $\mathbb{Z}^2$ -action contained in G_{disc} . Fundamental rectangles of size $a \times b$ then have symplectic area

$$\int_{\text{cell}} \omega = \int_0^a \int_0^b dp dq = ab.$$

By Theorem 3.4 we require $ab = 2\pi\hbar$ for a minimal cell.

4.2 Heisenberg-type relations on states

Let $\mathcal{H}_{\text{twin}}$ be the twin Hilbert space (constructed below). It is essential that the Heisenberg phase *cannot* arise if both translations are represented as naive coordinate shifts of phase-space functions, $\psi(q, p) \mapsto \psi(q - a, p)$ and $\psi(q, p) \mapsto \psi(q, p - b)$: such shifts commute identically, since both send $\psi(q, p) \mapsto \psi(q - a, p - b)$ irrespective of a, b . The phase requires the configuration *polarization* (functions of q alone), or equivalently the prequantum line-bundle connection whose holonomy around a cell is its enclosed symplectic area. In the configuration polarization on $L^2(\mathbb{R}_q)$ one sets

$$(T_q(a)\psi)(q) := \psi(q - a) = e^{-iaP/\hbar}\psi, \quad (T_p(b)\psi)(q) := e^{-ibq/\hbar}\psi(q) = e^{-ibQ/\hbar}\psi,$$

so that $T_q(a)$ shifts position while $T_p(b)$ is the momentum boost.

Traversing a minimal rectangle in the (q, p) -plane yields a phase:

Proposition 4.1 (Braiding of translations). *With the Weyl operators above,*

$$T_q(a)T_p(b) = e^{iab/\hbar}T_p(b)T_q(a). \quad (4.1)$$

In particular, when the cell area is an integer multiple of the minimal area, $ab \in 2\pi\hbar\mathbb{Z}$, the phase is $e^{iab/\hbar} = 1$ and the lattice translations $T_q(a), T_p(b)$ commute; a non-trivial phase $e^{iab/\hbar} \neq 1$ arises precisely for sub-cell (fractional) displacements.

To obtain non-trivial phases in the quantum regime one typically works with parameters (a, b) not corresponding to a single minimal cell but to fractional displacements relative to the minimal area scale. This leads to the Weyl form of the canonical commutation relations.

Theorem 4.2 (Geometric canonical commutation). *Let a, b be sufficiently small and let $T_q(a), T_p(b)$ be the corresponding unitary translations on $\mathcal{H}_{\text{twin}}$. Under the quantization condition, there exists $\hbar > 0$ such that*

$$T_q(a)T_p(b) = e^{iab/\hbar}T_p(b)T_q(a). \quad (4.2)$$

Formally differentiating with respect to a and b at 0 yields densely defined self-adjoint generators Q, P such that

$$[Q, P] = i\hbar\text{Id}$$

on a common invariant core.

Sketch of proof. The phase factor $e^{iab/\hbar}$ corresponds to the symplectic area enclosed by the loop obtained by composing the two translations in different orders. The discrete group action quantizes this area in units of $2\pi\hbar$. Relation (4.2) is the Weyl form of the canonical commutation relations. Expanding $T_q(a)$ and $T_p(b)$ as $T_q(a) = e^{-iaP/\hbar} \approx \text{Id} - iaP/\hbar$ and $T_p(b) = e^{-ibQ/\hbar} \approx \text{Id} - ibQ/\hbar$ for small a, b and comparing terms of order ab yields $[Q, P] = i\hbar\text{Id}$. $\square$

Remark 4.3. The construction extends to higher dimensions and to more general configuration manifolds, where the role of q, p is played by local canonical coordinates and the discrete group action is adapted to the underlying geometry.

5 Twin Hilbert spaces as L^2 on orbit space

We now construct the Hilbert space of the theory as an L^2 -space on the orbit space.

5.1 Invariant measures on the quotient

We assume:

- The action of G_{disc} on M is properly discontinuous and admits a fundamental domain $F \subset M$ of finite symplectic volume;
- The symplectic volume form $\Omega := \omega^{\wedge n}$ induces a G_{disc} -invariant measure on M .

The quotient $X = M/G_{\text{disc}}$ is then equipped with a natural measure μ defined by

$$\int_X f([x]) \, d\mu([x]) := \int_F f(\pi(x)) \, \Omega(x)$$

for reasonable test functions f .

Proposition 5.1. *The measure μ is well-defined (independent of the choice of F up to null sets) and σ -finite. If F has finite volume, then $\mu(X) < \infty$.*

Idea. The invariance of Ω under G_{disc} ensures consistency of the definition on overlapping translates of F . Proper discontinuity guarantees that only countably many translates intersect a given compact set, leading to σ -finiteness. $\square$

5.2 Construction of the twin Hilbert space

Definition 5.2 (Twin Hilbert space). The *twin Hilbert space* associated with $(M, \omega, G_{\text{disc}})$ is

$$\mathcal{H}_{\text{twin}} := L^2(X, \mu),$$

the space of (equivalence classes of) complex-valued, square-integrable functions on the orbit space $X = M/G_{\text{disc}}$.

Proposition 5.3. $\mathcal{H}_{\text{twin}}$ is a separable Hilbert space with inner product

$$\langle \psi, \varphi \rangle := \int_X \overline{\psi([x])} \, \varphi([x]) \, d\mu([x]).$$

Proof. Standard L^2 theory on a σ -finite measure space. Separability follows from the existence of a countable dense subset (e.g. simple functions with rational coefficients). $\square$

Remark 5.4 (Prequantum space vs. polarization). The space $L^2(X, \mu)$ built from the *full* phase space $X = M/G_{\text{disc}}$ is the *prequantum* Hilbert space. As in geometric quantization it is reducible—heuristically “twice too large”—and the irreducible physical Hilbert space is obtained by choosing a *polarization* (a maximal involutive Lagrangian distribution) and keeping only polarized sections. In the configuration polarization the physical space is L^2 of the configuration quotient, e.g. $L^2(\mathbb{R}_q)$ in Section 7. Throughout we regard $L^2(M/G_{\text{disc}})$ as the ambient prequantum space and the physical $\mathcal{H}_{\text{twin}}$ as its polarized subspace; the Weyl operators of Section 4 act in this polarization.

5.3 Representation of translations and observables

The discrete action and symplectic structure induce natural operators:

- For each $g \in G_{\text{disc}}$, a unitary operator $U(g)$ on $\mathcal{H}_{\text{twin}}$ defined by transport along the orbit; in many examples $U(g)$ is trivial at the quotient level but non-trivial when internal phases are included;
- Multiplication operators $(A\psi)([x]) = a([x])\psi([x])$ for suitable measurable functions $a : X \rightarrow \mathbb{R}, \mathbb{C}$, corresponding to position or other configuration observables;
- Differential or pseudo-differential operators representing momentum and Hamiltonians, obtained by lifting vector fields on M and projecting them to the orbit space.

Theorem 5.5 (Twin Hilbert construction). *Under the assumptions above, the triple $(\mathcal{H}_{\text{twin}}, U, \mathcal{A})$, where $\mathcal{A}$ is a suitable $*$ -algebra of operators generated by observables and translations, provides a representation of the twin quantum geometry compatible with the symplectic form and the discrete action.*

Remark 5.6. In later works, we refine $\mathcal{H}_{\text{twin}}$ by incorporating internal phases and cohomological data, leading to projective representations and central extensions of G_{disc} . This is the natural arena for the full Weyl system associated with (Q, P) .

6 Twin time evolution and Schrödinger dynamics

We now show how Schrödinger-type unitary evolution arises from covariant transport of Hamiltonian flows.

6.1 Hamiltonian flows on (M, ω)

Let $H_{\text{cl}} : M \rightarrow \mathbb{R}$ be a smooth Hamiltonian. The associated Hamiltonian vector field X_H is defined by

$$\iota_{X_H}\omega = dH_{\text{cl}}.$$

The flow $\Phi_t : M \rightarrow M$ generated by X_H preserves ω and hence the symplectic volume form Ω .

We assume that Φ_t commutes with the action of G_{disc} or, more generally, that it is compatible up to phase factors determined by the cohomological data.

6.2 Induced evolution on the orbit space

The flow Φ_t descends to a measurable flow

$$\tilde{\Phi}_t : X \rightarrow X, \quad \tilde{\Phi}_t([x]) = [\Phi_t(x)],$$

which preserves the measure μ .

Definition 6.1 (Twin time evolution). The induced unitary group $(U_t)_{t \in \mathbb{R}}$ on $\mathcal{H}_{\text{twin}}$ is defined by

$$(U_t\psi)([x]) := \psi(\tilde{\Phi}_{-t}([x])).$$

Proposition 6.2. *Each U_t is unitary on $\mathcal{H}_{\text{twin}}$, and $(U_t)_{t \in \mathbb{R}}$ is a strongly continuous one-parameter group of unitaries.*

Proof. Unitarity follows from invariance of μ under $\tilde{\Phi}_t$. Strong continuity is a consequence of the continuity of the flow and the dominated convergence theorem, for instance on a core of continuous compactly supported functions on X . $\square$

6.3 Generator and Schrödinger equation

By Stone's theorem there exists a self-adjoint operator $\hat{H}$ on $\mathcal{H}_{\text{twin}}$ such that

$$U_t = e^{-it\hat{H}/\hbar}.$$

Theorem 6.3 (Emergent Schrödinger dynamics). *Let $\psi_t = U_t\psi_0$ for some initial state $\psi_0 \in \mathcal{H}_{\text{twin}}$. Then ψ_t satisfies*

$$i\hbar \frac{\partial}{\partial t} \psi_t = \hat{H} \psi_t$$

in the sense of strongly continuous unitary evolution. The operator $\hat{H}$ is the twin-quantized Hamiltonian associated with H_{cl} and the discrete symplectic action.

Proof. This is a direct consequence of Stone's theorem. The essential point is that the dependence on $\hbar$ has already been fixed by the symplectic quantization condition of Section 3. The generator $\hat{H}$ is determined (up to additive constants) by the classical Hamiltonian and the chosen representation of translations and observables on $\mathcal{H}_{\text{twin}}$. $\square$

Remark 6.4 (Transport vs. Schrödinger generator). The unitary $U_t\psi = \psi \circ \tilde{\Phi}_{-t}$ is the *Koopman* operator of the classical flow; its generator on the prequantum space is the Liouville operator $\hat{H}_{\text{pre}} \sim -i\hbar X_H$, whose spectrum is the classical (Liouville) spectrum, *not* the quantum one. Recovering the Schrödinger operator—with its discrete spectrum and zero-point energy—requires the polarization of Remark 5.4 together with the prequantum connection term, i.e. the geometric-quantization operator $\hat{H} = -i\hbar X_H + (H_{\text{cl}} - \theta(X_H))$ restricted to polarized sections. The statements of this section are to be read in that polarized sense.

Remark 6.5. In explicit models (harmonic oscillator, free particle, particle on a circle), $\hat{H}$ recovers the familiar Schrödinger operators, with $\hbar$ and canonical commutation relations inherited from the twin geometry.

7 Examples and simple models

We work out three canonical examples, showing how the symplectic-area quantization of §3 reproduces the standard spectra; the sketches below are made fully precise in a later version.

7.1 Harmonic oscillator

Take $M = \mathbb{R}^2$ with coordinates (q, p) and

$$H_{\text{cl}}(q, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

Choose a rectangular lattice generated by translations $(q, p) \mapsto (q + a, p)$ and $(q, p) \mapsto (q, p + b)$ with $ab = 2\pi\hbar$. Fundamental twin cells have area $2\pi\hbar$. The twin Hilbert space $\mathcal{H}_{\text{twin}} = L^2(X, \mu)$ carries a representation of (Q, P) satisfying $[Q, P] = i\hbar \text{Id}$. The twin Hamiltonian $\hat{H}$ defined via the unitary group U_t reproduces, in the configuration polarization (Remark 5.4; cf. Remark 6.4), the usual spectrum $E_n = \hbar\omega(n + \frac{1}{2})$. Concretely the framework *fixes* this spectrum by area quantization. A classical orbit $H_{\text{cl}} = E$ is the ellipse $\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = E$, with semi-axes $\sqrt{2mE}$ (in p) and $\sqrt{2E/(m\omega^2)}$ (in q), hence enclosed symplectic area

$$\oint_{H=E} p \, dq = \pi \sqrt{2mE} \sqrt{\frac{2E}{m\omega^2}} = \frac{2\pi E}{\omega}.$$

Bohr–Sommerfeld quantization in units of the minimal twin cell, with the Maslov index $\frac{1}{2}$ supplied by the polarization (Remark 5.4), reads $\oint p \, dq = 2\pi\hbar(n + \frac{1}{2})$, i.e. $\frac{2\pi E}{\omega} = 2\pi\hbar(n + \frac{1}{2})$, giving exactly $E_n = \hbar\omega(n + \frac{1}{2})$. The zero-point energy $\frac{1}{2}\hbar\omega$ is the area of half a fundamental cell — a direct geometric reading of the paper’s central quantity $\Delta\mathcal{A}_{\min} = 2\pi\hbar$.

7.2 Free particle

For a free particle of mass m on the line, $H_{\text{cl}}(q, p) = p^2/(2m)$. The twin structure quantizes the phase space area and identifies $\hbar$ as the minimal cell area divided by 2π . The twin Hilbert space is $L^2(\mathbb{R}, dq)$ —the configuration polarization of the prequantum space $L^2(M/G_{\text{disc}})$ (Remark 5.4)—and the generator of translations in q acts as the usual momentum operator $P = -i\hbar\partial_q$.

7.3 Particle on a circle

Taking $M = T^*S^1 \simeq S^1 \times \mathbb{R}$ with angle coordinate θ and momentum p , the symplectic form is $\omega = dp \wedge d\theta$. Quantization of $\int \omega$ along cycles associated with $\theta \mapsto \theta + 2\pi$ yields discrete momentum eigenvalues $p_n = n\hbar$. This fits naturally in the twin framework, with G_{disc} generated by 2π -shifts in θ and integer shifts in p . Explicitly, in the configuration polarization the physical states are functions $\psi(\theta)$ on S^1 ; a momentum eigenstate $\psi(\theta) = e^{ip\theta/\hbar}$ is single-valued under $\theta \mapsto \theta + 2\pi$ iff $p \cdot 2\pi/\hbar \in 2\pi\mathbb{Z}$, i.e. $p_n = n\hbar$ with $n \in \mathbb{Z}$. Equivalently, the holonomy $\exp(\frac{i}{\hbar} \oint p \, d\theta) = 1$ around the θ -cycle is the integrality condition of §3, and the discrete spectrum $p_n = n\hbar$ is its solution — the simplest instance of emergent quantization in the twin framework.

Conclusion

We have shown how the Planck constant, canonical commutation relations, Hilbert space structure and Schrödinger evolution can be derived from a purely geometric framework based on discrete symplectic group actions on phase space. This provides a rigorous and conceptually unified foundation for the twin formulation of quantum mechanics. In subsequent articles we will extend this construction to a full operator and measurement theory, and further to quantum fields, kernels and gravity-like effects.